MANAGEMENT INFORMATION SYSTEM

UNIT – II

Information Systems for Decision Making

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Unit at a glance

Topics covered (as per syllabus):

1. Evolution of an Information System **2.** Basic Information Systems **3.** Decision Making & MIS **4.** Types of Decisions – Structured vs Unstructured **5.** Strategic, Tactical & Operational Information **6.** Simon's Model of Decision-Making **7.** MIS as a Technique for Programmed Decisions **8.** Decision Assisting Information Systems

1 Evolution of an Information System

Information systems did not appear overnight. They grew step by step – from handwritten registers to AI-powered systems – as organisations needed faster, more accurate and more useful information for decisions.

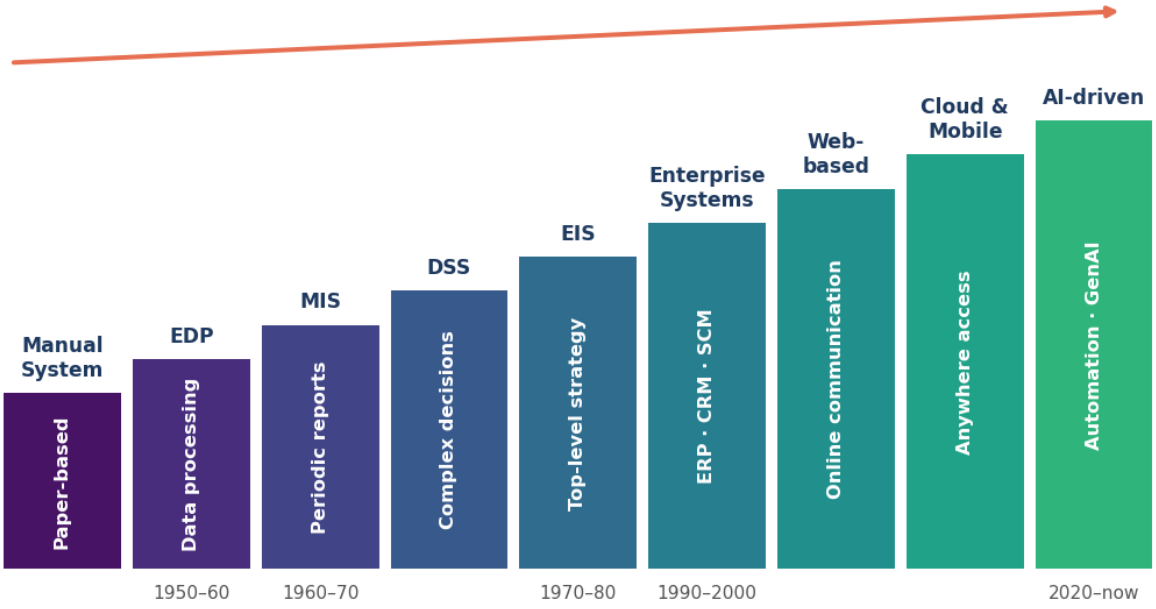


Fig 1: Evolution of information systems (staircase of growth)

Stage	Period	What it did	Key features
1. Manual System	Before computers	Records kept by hand in registers and files	Slow, error-prone, hard to search or share
2. EDP (Electronic Data Processing)	1950 – 1960	Computers used for routine data processing (payroll, billing)	Transition from manual work; automates clerical tasks
3. MIS	1960 – 1970	Produces periodic, summarised reports for managers	Reports to managers for control and planning

Stage	Period	What it did	Key features
4. DSS (Decision Support System)	After MIS	Helps with complex decisions using models and data	Makes complex decision-making easier; interactive
5. EIS (Executive Information System)	1970 – 1980	Gives top-level executives quick summary information	Top-level, strategy making; sensory / group support
6. Enterprise Systems	1990 – 2000	Integrates all departments on one platform	ERP, CRM, SCM; covers every business function
7. Web-based System	After 2000	Online systems accessible via internet	Online communication, e-commerce, sharing
8. Cloud & Mobile System	Recent years	Data and apps stored on cloud, used on phones	Information accessible anywhere, anytime
9. AI-driven System	2020 – present	AI/GenAI automates analysis and decisions	Automation, prediction, generative AI

Remember

EDP → MIS → DSS → EIS → ERP → Web → Cloud/Mobile → AI. Each stage solved the limitation of the previous one: EDP processed data, MIS reported it, DSS supported decisions, EIS served top management, ERP integrated everything, and AI now automates and predicts.

2 Basic Information Systems

An **information system (IS)** is a set of people, data, processes and technology that collects, processes, stores and distributes information to support operations, management and decision-making. The basic types used in organisations are:

System	Full form	Used by	Purpose
TPS	Transaction Processing System	Operational staff	Records day-to-day transactions (sales, billing, payroll)
MIS	Management Information System	Middle managers	Summary & exception reports for control and planning
DSS	Decision Support System	Middle / senior managers	Analyses data and models to support semi-structured decisions
EIS / ESS	Executive Information / Support System	Top management	Dashboards of key indicators for strategic decisions
Expert System	Knowledge-based system	Specialists / managers	Imitates an expert to give advice in a specific domain

3 Decision Making & MIS

3.1 What is Decision Making?

Decision making is the process of choosing the best course of action from several alternatives to achieve a goal. Good decisions need **accurate, relevant and timely information** – and supplying that is the main job of MIS.

3.2 Role of MIS in Decision Making

1. **Provides relevant information** – the right data in the right form.
2. **Improves decision quality** – decisions are based on facts, not guesses.
3. **Identifies problems** – shows deviations and warning signs early.
4. **Supports planning** – gives data for short- and long-term plans.
5. **Helps compare alternatives** – lets managers weigh options side by side.
6. **Provides timely information** – reports arrive when they are needed.
7. **Supports monitoring & controlling** – tracks actual vs planned performance.
8. **Reduces uncertainty** – forecasts and analysis lower the risk.
9. **Improves coordination** – all departments work on the same information.
10. **Saves time & money** – automation cuts manual effort and wasteful decisions.

3.3 What MIS Does for the Decision Maker

Activity	Meaning
Data collection	Gathers raw facts from internal and external sources
Data processing	Sorts, calculates and organises the data
Information analysis	Finds patterns and trends in processed data
Report generation	Creates regular, summary and exception reports
Forecasting	Predicts future sales, demand, costs
What-if analysis	Tests how a change (price, cost) would affect results
Problem identification	Spots issues such as falling sales or rising cost
Evaluation	Judges alternatives against the objectives
Monitoring & controlling	Compares performance with targets and corrects gaps
Strategy planning	Supports long-term direction of the organisation

4 Types of Decisions

Decisions can be classified in three ways – by their **nature**, by their **structure**, and by the **management level** that takes them.

4.1 Based on Nature

Programmed decisions	Non-programmed decisions
Routine and repetitive	New, unusual, one-time
Solved with fixed rules/procedures	No ready-made rules; needs judgement
Low risk; taken by lower management	High risk; taken by top management
Example: reordering stock when it falls below a limit	Example: entering a new market or launching a new product

4.2 Based on Structure – Structured vs Unstructured

Basis	Structured	Semi-structured	Unstructured
Meaning	Clear procedure and rules exist	Part of the problem has rules, part needs judgement	No rules; problem is vague
Information needed	Exact & available	Partly available	Incomplete / uncertain
Level	Operational	Tactical (middle)	Strategic (top)
Supported by	TPS, MIS	DSS	EIS, Expert systems
Example	Calculating salary	Preparing a quarterly budget	Deciding a 5-year business strategy

Link to remember

Programmed ≈ Structured ≈ Routine | Non-programmed ≈ Unstructured ≈ Unique

5 Strategic, Tactical & Operational Information

Every level of management needs a different kind of information for taking decisions.



Fig 2: Management levels and the decisions taken at each

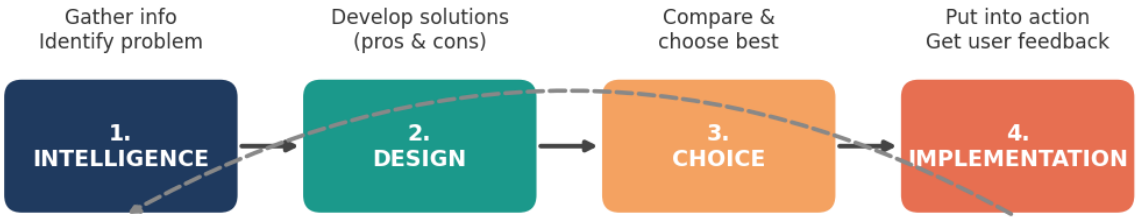
Feature	Strategic information	Tactical information	Operational information
Used by	Top management	Middle management	Lower / first-line management
Decision type	Unstructured	Semi-structured	Structured
Time horizon	Long term (years)	Medium term (months)	Short term (daily / weekly)
Detail level	Highly summarised	Moderately detailed	Very detailed & complex
Source	Mostly external + internal	Mostly internal	Internal transactions
Purpose	Set goals & overall direction	Allocate resources, implement plans	Carry out day-to-day tasks

Feature	Strategic information	Tactical information	Operational information
Example	Expanding to a new country	Monthly budget for a department	Today's stock level, attendance

- **Tactical information** – supports decisions that follow a structure or are semi-structured; it lies at the middle level and links strategy to daily work.
- **Operational information** – detailed and complex data from the operational (last) level; decisions here are structured.

6 Simon's Model of Decision-Making

Herbert A. Simon described decision-making as a **four-phase process**. MIS supports each phase.



Feedback loop – problems found go back to Intelligence

Fig 3: Simon's decision-making model

Phase	What happens	Role of MIS
1. Intelligence	Gather information and identify the problem	Collects data, shows deviations and signals the problem
2. Design	Develop possible solutions, list advantages & disadvantages, prepare a report	Gives models, forecasts and what-if analysis to build alternatives
3. Choice	Compare the solutions and choose the best one	Provides comparison reports so the best option can be selected
4. Implementation	Put the solution into action – give it to the user, take user feedback	Monitors results, gives feedback and corrects the plan

7 MIS as a Technique for Making Programmed Decisions

Programmed decisions are routine decisions made by applying **predefined rules**. MIS is ideal for them because computers can follow rules quickly and without mistakes.

7.1 How MIS Helps in Programmed Decision Making

1. **Collects routine data** automatically from daily operations.
2. **Processes data automatically** without manual work.
3. **Applies predefined rules** (e.g., "if stock < 50, reorder").
4. **Generates regular reports** on a fixed schedule.

5. **Identifies deviations** from the normal or planned level.
6. **Supports quick decisions** because rules are already set.
7. **Reduces human error and effort.**

7.2 Flow of a Programmed Decision

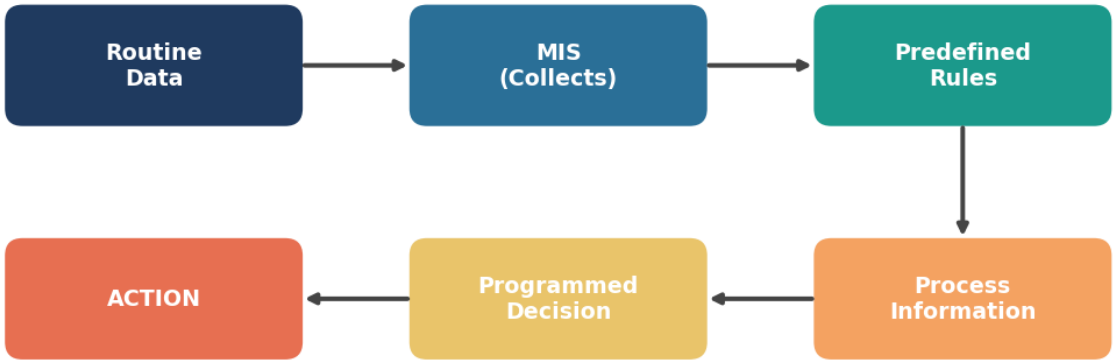


Fig 4: Programmed decision flow using MIS

> **Example – Inventory reorder**
> Sales data (routine data) → MIS collects it → rule: "reorder when stock falls below 50 units" → MIS processes stock level → decision: place order → action: purchase order sent to the supplier.

8 Decision Assisting Information Systems (DAIS)

Decision Assisting Information Systems help managers with decisions that cannot be handled by fixed rules alone – mainly semi-structured and unstructured ones. They do not replace the manager; they give analysis, models and options so that a better decision can be made.

8.1 Types of DAIS

Type	Full form	Key idea
DSS	Decision Support System	Interactive system using data and models to analyse alternatives for semi-structured problems
EIS / ESS	Executive Information / Support System	Easy, graphical summaries of key performance indicators for top management
Expert System	Knowledge-based system	Stores expert knowledge and rules to give advice like a human specialist
GDSS	Group Decision Support System	Supports a team taking a decision together – brainstorming, voting, discussion

8.2 How a DAIS Works

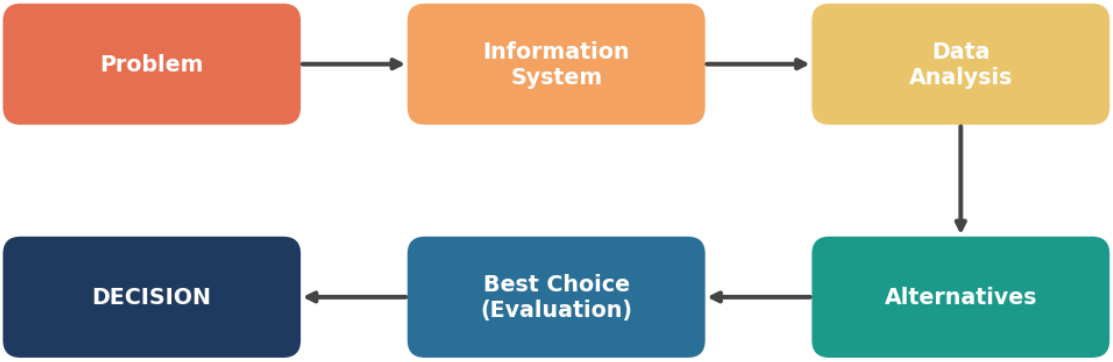


Fig 5: Working of a decision assisting information system

8.3 Importance of DAIS

- **Complex problem solving** – handles problems with many variables.
- **Strategic problem handling** – supports long-term, high-impact decisions.
- **Better decision rules** – compares alternatives and picks the best one with evidence.
- **Speed and accuracy** – analysis that would take days is done in minutes.
- **Group support** – improves teamwork when many people decide together.

8.4 MIS vs DSS (quick comparison)

Basis	MIS	DSS
Decision type	Structured / programmed	Semi-structured / complex
Output	Fixed, periodic reports	Flexible, on-demand analysis
User role	Receives reports	Interacts with models and data
Focus	Efficiency & control	Effectiveness & problem solving

★ Quick Revision & Exam Questions

🔑 Key points in one glance

• Evolution: Manual → EDP → MIS → DSS → EIS → Enterprise (ERP/CRM/SCM) → Web → Cloud & Mobile → AI-driven.

• Decisions: Programmed vs Non-programmed; Structured, Semi-structured, Unstructured; Strategic, Tactical, Operational.

• Simon's model: **Intelligence → Design → Choice → Implementation** (+ feedback).

• MIS handles **programmed** decisions using predefined rules; DAIS (DSS, EIS/ESS, Expert, GDSS) handles complex ones.

Likely questions

1. Explain the evolution of information systems with a diagram.
2. Describe the role of MIS in decision making.

3. Differentiate between structured and unstructured decisions.
4. Explain strategic, tactical and operational information with examples.
5. Explain Simon's model of decision-making. How does MIS support each phase?
6. How does MIS help in programmed decision making? Explain with a flow diagram.
7. What are decision assisting information systems? Explain their types and importance.